

From local payoffs to global instabilities: A spectral cartography of spatiotemporal chaos in canonical 2×2 evolutionary games

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(Received 5 March 2026; accepted 23 July 2026; published 7 August 2026)

We develop a motif-based framework for spatiotemporal chaos in spatial evolutionary games and use it to map the dynamical phase diagram in the payoff plane. Using Boolean linearization of the imitate-the-best rule, we derive analytical instability thresholds for local motifs including invaders, cooperative pairs, stripe interfaces, and cooperative cores. These thresholds are obtained from payoff balance at contested motif interfaces and recover classical invasion thresholds of spatial evolutionary games, which emerge here as boundaries of the chaotic phase. Combining the Derrida slope with the asymptotic Hamming distance, we obtain a four-region cartography: ordered, transient-chaotic, sustained-chaotic, and subcritical-chaotic dynamics. The phase diagram is organized by density-dependent motif selection: Different initial cooperator densities activate different instability mechanisms, yet a small set of motif-instability lines consistently bounds the sustained-chaos region across densities. This cartography reveals a subcritical chaotic phase (Derrida slope $s < 1$ but asymptotic Hamming distance $d_\infty > 0$), where infinitesimal perturbations decay while finite-amplitude perturbations sustain chaos. The motif-based framework is anchored by an exact benchmark: For homogeneous backgrounds, the Boolean Jacobian yields an exact correspondence between the Derrida slope and spectral radius, linking damage spreading to deterministic instability.

DOI: [10.1103/ytzb-pk8g](https://doi.org/10.1103/ytzb-pk8g)

I. INTRODUCTION

Nowak and May [1] observed that the deterministic “imitate-the-best” update rule generates chaotic dynamics in the spatial Prisoner’s Dilemma game. Simple local interactions give rise to complex and persistent spatial patterns, allowing cooperators and defectors to coexist through the constant formation and breakup of clusters. This result sparked broad interest in how spatial structure acts as a mechanism for sustaining cooperation. A large body of work has since examined spatial evolutionary games on lattices and complex networks, focusing on how interaction topology shapes asymptotic cooperator frequencies [2–5]; for comprehensive reviews of spatial social dilemmas from a statistical-physics perspective, see Refs. [6,7]. Yet while spatial pattern formation and phase transitions between strategy configurations are by now well reviewed, the chaotic dynamical richness identified by Nowak and May—transient structures, sensitivity to initial conditions, and irregular spatiotemporal behavior—has received comparatively less attention. At the population level, the mean-field counterpart of these models (e.g., the replicator equations) is well understood dynamically [8], but it smooths away the spatial heterogeneity that drives the phenomena of interest here.

Many dynamical models of evolutionary games arise as large-population limits of microscopic update rules, giving rise either to well-mixed ODEs or to spatially extended lattice and continuum formulations capable of pattern formation

and instability [8–12]. These models, including discrete ones, represent real-world collective-action problems such as social cooperation, public goods provisioning, and epidemiological spread [13–15]. Understanding when such models develop instability and emergent complexity requires methods that operate directly at the level of the microscopic update rule.

The present paper addresses this gap through a unifying perspective: The onset of macroscopic damage spreading in spatial evolutionary games is controlled by the loss of stability of characteristic local spatial motifs. Each instability threshold is set by a simple condition: The two strategies competing at the motif’s interface earn equal payoffs. Different motifs generate different instability mechanisms and therefore different phase boundaries in the payoff plane. The resulting motif-instability structure organizes the observed chaotic regions and explains their dependence on the initial cooperator density.

Two complementary analytical traditions address stability in dynamical systems. The first is Fourier stability analysis, which linearizes the governing dynamics and diagnoses spatial instabilities through the growth of plane-wave perturbations; it is the standard tool for continuum systems and coupled-map lattices [12,16,17]. The second is damage spreading, introduced by Derrida and Pomeau [18], which quantifies chaotic sensitivity in discrete symbolic systems through the probabilistic growth of a small perturbation between two initially near-identical replicas. The present paper connects these two perspectives in the setting of spatial evolutionary games. The linear stability of small characteristic spatial patterns—the *motifs* defined in Sec. IV—controls the

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onset of macroscopic damage spreading, and hence the location of the chaotic phase boundaries in the payoff plane.

Alfaro and Sanjuán [19] recently established the normalized Hamming distance as a reliable numerical indicator of spatial chaos in evolutionary games, saturating within the chaotic regimes originally observed by Nowak and May and vanishing otherwise. While their work provides a robust numerical characterization, an analytical framework linking the onset of spatial chaos to the microscopic update dynamics has remained unavailable.

A fundamental obstacle to linking microscopic update rules with macroscopic instability is the discrete and nondifferentiable nature of cellular-automaton dynamics. Because each update selects the maximum among the finitely many payoffs in an agent's neighborhood, rather than applying a smooth transformation of the state, conventional Jacobian-based linearization is unavailable. We overcome this by working in the sparse-damage limit, where the initially perturbed sites are sufficiently rare that they evolve independently. In this regime the Boolean derivative [20,21] yields an exact linear operator governing first-order perturbation propagation, providing a spectral description of instability in an otherwise nonsmooth system.

For a homogeneous reference background, translational invariance renders the Boolean Jacobian a block-circulant convolution operator, exactly diagonalized by the discrete Fourier transform; its spectral radius $\Lambda = \max_{\mathbf{k}} |\lambda(\mathbf{k})|$ coincides exactly with the Derrida slope ($s = \Lambda$). In this exactly solvable case, damage spreading and spectral instability coincide. The Derrida contours mapped in Sec. III, however, are obtained from random initial conditions, where the lattice forms a mosaic of locally ordered motifs (uniform regions, stripes, and cooperative cores). Within this motif-based effective-medium picture, the onset of damage spreading is associated with the first statistically prevalent motif whose linearized dynamics becomes marginally unstable ($\Lambda_M = 1$). Our computations further reveal a subcritical regime ($s < 1$, $d_\infty > 0$) in which finite-amplitude perturbations sustain chaos despite linear stability.

The critical condition $s \approx 1$ can be expressed analytically through payoff-balance relations, one for each spatial motif. This criterion analytically predicts the boundaries of the spatiotemporal chaos regions originally observed through numerical simulation by Nowak and May [1,22] and supplies a unifying spectral interpretation of the discrete cluster-invasion thresholds derived in later literature [3,4,23]. The connection between these thresholds and chaotic phase boundaries had not previously been established.

Combining the Derrida slope with the asymptotic Hamming distance d_∞ [19], we construct a spectral cartography of the full (u, v) payoff plane distinguishing ordered behavior, transient chaos, sustained chaos, and subcritical chaos across multiple initial cooperator densities. The main contributions of this work are therefore the following:

(1) *Motif-controlled chaotic cartography.* The four-regime phase diagram is organized by the full set of motif-instability lines, with the initial cooperator density selecting which motifs are active. Within this organization a small, fixed subset (notably lines III, V, and VIII) bounds the *sustained*-chaos region at every density studied, so the topology of the phase

diagram is largely invariant with cooperator density rather than fragmenting into density-specific cases.

(2) *A subcritical chaotic phase.* We identify and map a regime with $s < 1$ yet $d_\infty > 0$: Infinitesimal perturbations decay, but finite-amplitude perturbations sustain chaos, so the ordered state is metastable. This is a dynamical phase distinct from linear instability, not a reinterpretation of any invasion threshold.

(3) *Analytical prediction of phase boundaries.* Closed-form motif marginal-stability conditions—obtained from payoff balance at each motif's contested interface—reproduce the known cluster-invasion thresholds and are identified, for the first time, as the boundaries of the chaotic phase.

(4) *A Boolean spectral framework.* We connect probabilistic damage spreading, the spectral stability of the Boolean Jacobian, and motif geometry. For homogeneous backgrounds the connection is exact: The Derrida slope equals the spectral radius, $s = \Lambda$.

The paper is organized as follows. Section II introduces the model. Section III presents the numerical chaos measures. Section IV develops the analytical framework. Section V classifies the dynamical regimes. Section VI concludes.

II. CANONICAL 2×2 GAMES AND THE MODEL

We study the deterministic cellular automaton introduced by Nowak and May [1,24], in which *canonical 2×2 games* are played on a square lattice of size $L \times L$. Each site s_{ij} is occupied by an individual who adopts one of two strategies, cooperator (C) or defector (D), and plays against their four von Neumann neighbors. Payoffs are determined by the two-parameter matrix for canonical 2×2 games [25]:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix} = \begin{pmatrix} 1 & v \\ 1+u & 0 \end{pmatrix}, \quad (1)$$

where $u = T - R$ measures the *temptation to defect* and $v = S - P$ the *risk of cooperation*. Restricting to $-1 \leq u, v \leq 1$ is sufficient to capture the four canonical game families [26], whose defining regions partition the payoff plane as shown in Fig. 1(c):

(1) *Prisoner's Dilemma* ($u > 0, v < 0$): defection strictly dominates cooperation;

(2) *Snowdrift and Hawk-Dove* ($u > 0, v > 0$): a weak dilemma in which cooperators and defectors can coexist;

(3) *Stag Hunt* ($u < 0, v < 0$): a coordination dilemma in which mutual cooperation is optimal but fragile against strategic uncertainty;

(4) *Harmony or Invisible Hand* ($u < 0, v > 0$): cooperation strictly dominates, so no social dilemma arises [27,28].

These outcomes are valid for well-mixed populations [25]. Spatial structure can alter them substantially—sometimes sustaining cooperation where it would collapse in a well-mixed setting, and sometimes generating persistent chaotic dynamics. We study these spatial effects below.

A. Synchronous imitate-the-best rule

At each discrete time step $t \in \mathbb{N}$, every individual simultaneously computes their own payoff and that of each neighbor and then adopts the strategy of whoever scored

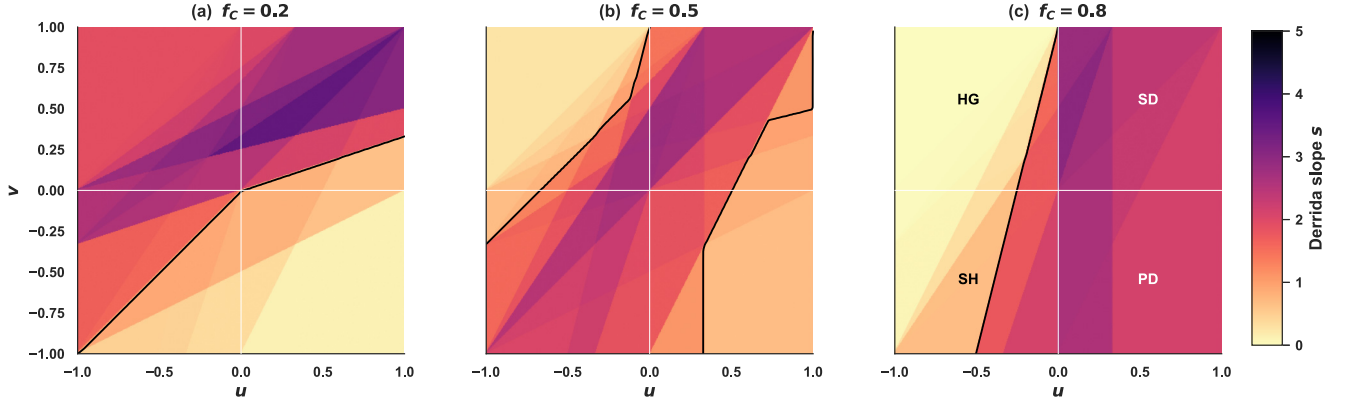


FIG. 1. The chaotic phase boundary shifts systematically with initial cooperator density, implying distinct dominant instability mechanisms at each density. Each panel shows the Derrida slope $s(u, v)$ for a fixed initial cooperator frequency f_c ; panels (a), (b), and (c) correspond to $f_c = 0.2, 0.5$, and 0.8 , respectively. Darker colors indicate $s > 1$ (chaotic phase) and lighter colors $s < 1$ (ordered phase). Black contours mark the empirical chaotic boundaries $s \approx 1$. The shape and position of these contours shift substantially with f_c (see, e.g., “chaotic corridor” appearing only at $f_c = 0.5$) indicating that the dominant instability mechanism depends on the initial spatial composition of the population. Section IV identifies which motif (and its corresponding spectral label) is responsible at each boundary. The canonical game regions—Harmony (HG), Stag Hunt (SH), Prisoner’s Dilemma (PD), and Snowdrift (SD)—are annotated in the rightmost panel for reference.

highest, retaining their current strategy in case of a tie. This synchronous “imitate-the-best” rule defines a nonlinear CA whose linearized dynamics we analyze in Sec. IV. We focus on synchronous updates, whose discrete translational invariance underlies the spectral analysis of Sec. IV; possible extensions to asynchronous and probabilistic rules are discussed in Sec. VI.

B. Payoff functions

For a focal individual with $n_C \in \{0, 1, 2, 3, 4\}$ cooperative neighbors, the payoffs under each strategy are as follows:

$$\Pi_C(n_C) = n_C + (4 - n_C)v, \quad (2)$$

$$\Pi_D(n_C) = n_C(1 + u). \quad (3)$$

Under the imitate-the-best rule, each agent compares its own payoff with those of its four neighbors and adopts the strategy of the highest scorer. At the contested interface of a spatial motif, the decisive comparison is therefore between a cooperator-defector pair, with each agent’s payoff evaluated at its *own* cooperative-neighbor count: $\Pi_C(n_C)$ for the cooperator versus $\Pi_D(n'_C)$ for the defector, where n_C and n'_C generally differ. Equating these two payoffs yields the payoff-balance condition of the motif (Table I).

III. QUANTIFYING CHAOS: HAMMING DISTANCE, DAMAGE SPREADING, AND THE DERRIDA SLOPE

Most studies of the spatial games track asymptotic cooperator frequencies [4,23]. We instead ask whether the dynamics is ordered or chaotic, quantifying sensitivity to initial conditions through damage spreading [18,29,30].

The damage-spreading protocol works as follows. A reference configuration A_0 is initialized randomly for a given (u, v) pair, and a perturbed copy B_0 is obtained by flipping the strategies at a small number of randomly chosen sites,

introducing initial damage d_0 . Both replicas are then evolved for one time step under the identical imitate-the-best rule $\mathcal{F}$: $A_1 = \mathcal{F}(A_0)$, $B_1 = \mathcal{F}(B_0)$. The resulting damage d_1 is measured by the normalized Hamming distance.

Definition 1 (Hamming distance). For two configurations A, B on the $L \times L$ lattice, the normalized Hamming distance is the fraction of sites at which their strategies differ:

$$d(A, B) = \frac{1}{L^2} \sum_{n,m=1}^L \mathbf{1}\{A(n, m) \neq B(n, m)\}, \quad (4)$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. We write $d_t := d(A_t, B_t)$ for the damage at time t .

Repeating this protocol over many initial-condition pairs gives the Derrida map, whose slope at the origin quantifies the *average one-step amplification* of an infinitesimal perturbation.

Definition 2 (Derrida map and slope). The Derrida map $g(d_0) = \mathbb{E}[d_1 | d(A_0, B_0) = d_0]$ is the expected postupdate damage averaged over an ensemble of configuration pairs with the same initial damage d_0 . The Derrida slope is its derivative at the origin:

$$s = \left. \frac{dg(d_0)}{dd_0} \right|_{d_0 \rightarrow 0}. \quad (5)$$

The Derrida slope s is the discrete-system analog of the largest Lyapunov multiplier: It measures the average per-step amplification of an infinitesimal perturbation in a symbolic system like a CA [21,31]. Thus $s > 1$ signals exponential growth of infinitesimal damage (chaotic phase), $s < 1$ contraction (ordered phase), and $s = 1$ the critical boundary. Because the dynamics is Boolean, no tangent space exists and a Lyapunov exponent cannot be defined in the usual way; the Derrida slope is the standard substitute for such symbolic systems, defined through the ensemble average of Definition 2. The connection between s and the spectral properties of the linearized rule is the subject of Sec. IV. CA can

TABLE I. The eight analytically derived instability lines that constitute the chaotic phase boundaries in the (u, v) payoff plane. Each line is the locus at which the named motif transitions from stable to unstable under the imitate-the-best rule. The dominant active line at a given initial cooperator density f_C determines the chaotic phase boundary observed numerically in Fig. 3; which line dominates depends on which local geometric motif is statistically prevalent at that density (see Sec. IV C).

Label	Critical line	Spectral label ($\mathbf{k}$)	Geometric configuration	Microscopic mechanism
I	$v = 4u + 1$	Long-wavelength (0,0)	Single D in all-C	Defector invasion
II	$v = (u + 1)/4$	Long-wavelength (0,0)	Single C in all-D	Cooperator invasion
III	$v = u/3$	Long-wavelength (0,0)	1D filament or pair	Nucleation threshold
IV	$v = u$	Stripe $(\pi, 0)/(0, \pi)$	Self-dual stripe	Domain-wall equilibrium
V	$v = 2u - 1$	Stripe $(\pi, 0)/(0, \pi)$	Cooperator kink	Transverse propagation
VI	$v = (u - 1)/2$	Stripe $(\pi, 0)/(0, \pi)$	Defector kink	Domain roughening
VII	$v = u + 2/3$	Oblique $(\pi, \pi/2)$	Oblique interface	Oblique instability
VIII	$u = 1/3$	Core $(0, \pi/3)$	Pocket D vs core C	Core erosion

exhibit mixed behavior in which ordered and chaotic regions coexist in the same rule [32,33]; here we find that the location and extent of the chaotic region depend not only on the payoff parameters (u, v) but also on the initial cooperator density $f_C(0)$, which sets the prevalence of the local motifs that drive instability.

We scan the (u, v) plane on 50×50 lattices [34] for $f_C(0) \in \{0.2, 0.5, 0.8\}$, averaging s over 30 independent initial conditions at each parameter point. Figure 1 summarizes the results and reveals two findings that motivate the framework developed in Sec. IV. First, the empirical chaotic phase boundaries (black contours, $s \approx 1$) are sharp and approximately linear in the (u, v) plane, suggesting that each boundary is governed by a single dominant instability mechanism. Second, both the location and the extent of the chaotic phase depend strongly on the initial cooperator frequency $f_C(0)$: A “chaotic corridor” spanning much of the SD and SH regions appears only at $f_C(0) = 0.5$, while high and low cooperator densities yield different chaotic regions with qualitatively different boundary geometries. The central question is therefore Which spatial motif becomes unstable at each boundary, and why does the dominant motif change with $f_C(0)$? Section IV answers this question.

IV. A UNIFIED ANALYTICAL FRAMEWORK FOR STABILITY BOUNDARIES

The numerical results of Sec. III establish that the Derrida slope s transitions sharply across approximately linear boundaries in the (u, v) plane and that the location of those boundaries depends on the initial cooperator density. The Derrida slope, however, measures only the *average* growth of a spatially random perturbation: It tells us *that* a transition occurs but not *which feature of the local update rule* is responsible.

To identify the instability mechanism, we shift from random perturbations to structured ones. Specifically, for an ordered reference background we ask how a plane-wave perturbation of wave vector $\mathbf{k}$ is amplified under a single application of the linearized update rule—a similar question asked of PDEs via linear stability analysis and of coupled-map lattices via Fourier decomposition [17,35]. The key insight is that particular instability mechanisms can be associated with and cooperative cores; see Fig. 2) together with their interface

perturbations such as stripe kinks. Each ordered background is naturally associated with a Fourier or Bloch decomposition, and the marginal-stability condition of that motif translates directly into a payoff-balance relation between the two competing strategies at the motif’s characteristic local geometry. The chaotic phase boundaries observed numerically are then the lines in the (u, v) plane at which one of these balance conditions is first satisfied—that is, at which the corresponding motif becomes marginally unstable.

The section is organized as follows. We first derive the Fourier-mode linearization and establish the link between the Derrida slope and the spectral radius (Sec. IV A). We then derive each motif’s payoff-balance line (Sec. IV B). Finally, we show which motif dominates at each initial cooperator density and compare the predicted lines with the numerical boundaries (Sec. IV C).

A. General formalism: Linearization and modal decomposition

1. Linearization of the update rule

Let $\delta_{n,m}(t) \in \{0, 1\}$ denote the perturbation field: $\delta_{n,m} = 1$ if site (n, m) differs between the two replicas (Definition 1). When the initial damage is sparse, damaged sites are separated by several lattice spacings and evolve independently. Perturbation propagation is then governed to leading order by the Boolean Jacobian of the update rule [20,21,31],

$$\delta_{n,m}(t+1) = \sum_{(r,s) \in \mathcal{N}_0} J_{n,m}(r,s) \delta_{n+r,m+s}(t), \quad (6)$$

where $\mathcal{N}_0$ contains the focal site and its von Neumann neighbors, and the Boolean derivative $J_{n,m}(r,s) = \partial s_{n,m}(t+1)/\partial s_{n+r,m+s}(t) \in \{0, 1\}$ equals 1 if flipping the state at offset (r, s) changes the next state of (n, m) . Because it is set by the local payoff ordering, $J_{n,m}(r,s)$ generally varies from site to site; its structure depends on the reference background.

We adopt a strict von Neumann neighborhood for both payoff accumulation and imitation, whereas Nowak and May [1] mixed von Neumann and Moore neighborhoods. Since the analysis below relies only on discrete translational invariance, it extends to Moore or other symmetric neighborhoods with the quantitative thresholds shifting accordingly.

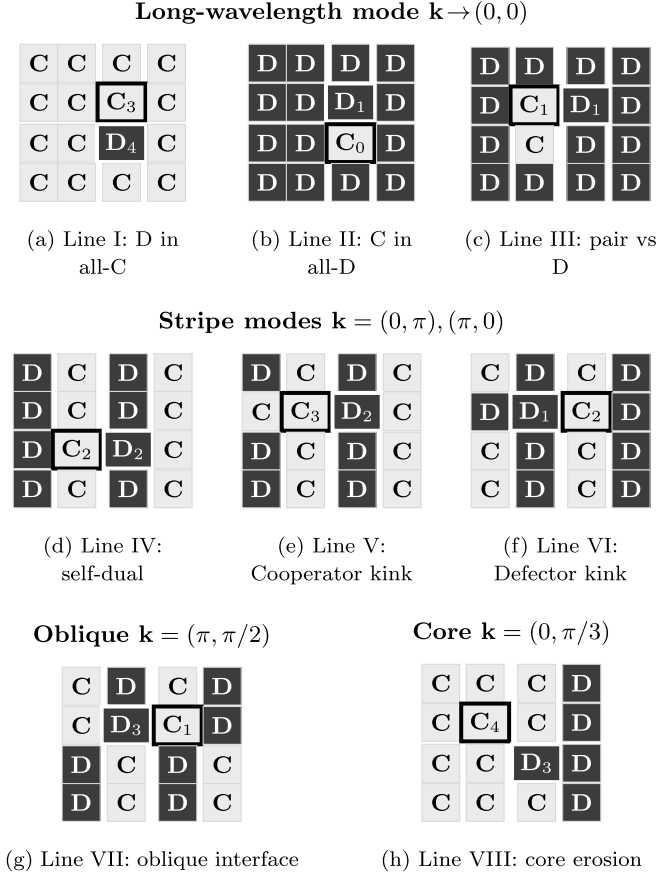


FIG. 2. Each chaotic phase boundary traces to a single microscopic competition event, fully determined by the payoff functions. Each panel shows the local configuration (motif) associated with the labeled sector; light cells are cooperators (C), dark cells are defectors (D). Cells with subscripted letters mark the critical competing pair whose payoff balance defines the motif's marginal-stability condition; subscripts give each agent's cooperative-neighbor count n_C . Panels (a)–(c) show that the long-wavelength limit $\mathbf{k} \rightarrow (0, 0)$ captures localized invasion geometries (single invader or two-site cluster); panels (d)–(f) probe domain-wall stability and its transverse kink instabilities; panels (g)–(h) correspond to higher-wavenumber instabilities that break the stripe symmetry or erode cooperative cores.

2. Homogeneous backgrounds: Dispersion relation and the exact identity $s = \Lambda$

When every site holds the same strategy, the payoff ordering—and hence the Boolean derivative—is identical at every site, and the Jacobian reduces to a scalar convolution,

$$\delta_{n,m}(t+1) = J_0 \delta_{n,m}(t) + J_* \sum_{(n',m') \in \mathcal{N}_{n,m}} \delta_{n',m'}(t). \quad (7)$$

Because the background is uniform and the rule isotropic, the sensitivity J_* is the same for all four neighbors; this is the only setting in which that is guaranteed. Substituting the plane-wave ansatz $\delta_{n,m}(t) = \tilde{\delta}_{\mathbf{k}}(t) e^{i(k_x n + k_y m)}$, the focal site contributes J_0 and the four neighbors contribute $2J_*(\cos k_x + \cos k_y)$, so the modes decouple:

$$\begin{aligned} \tilde{\delta}_{\mathbf{k}}(t+1) &= \lambda(\mathbf{k}) \tilde{\delta}_{\mathbf{k}}(t), \\ \lambda(\mathbf{k}) &= J_0 + 2J_*[\cos(k_x) + \cos(k_y)]. \end{aligned} \quad (8)$$

The linearized dynamics is stable if and only if the spectral radius $\Lambda = \max_{\mathbf{k}} |\lambda(\mathbf{k})| \leq 1$.

For this background the Derrida slope coincides exactly with the spectral radius,

$$s = \Lambda, \quad (9)$$

even though the former is an average over all modes and the latter a maximum. The two agree because the Boolean coefficients are idempotent ($J_0^2 = J_0, J_*^2 = J_*$): The Brillouin-zone average of $|\lambda(\mathbf{k})|^2$ equals $J_0^2 + 4J_*^2 = J_0 + 4J_*$, which is also $\max_{\mathbf{k}} |\lambda(\mathbf{k})|$ (Appendix A 1). The identity (9) is exact and anchors the framework: In the solvable homogeneous case, damage spreading and spectral instability are the same quantity.

3. Periodic and random backgrounds

For a background with a nontrivial unit cell—a stripe or a cooperative core—the Boolean derivative differs between site classes within the cell. The Jacobian is then circulant at the level of unit cells (block-circulant) and is diagonalized by a Bloch-Fourier transform, with marginal stability given by $\rho(\hat{J}(\mathbf{k})) = 1$, where $\hat{J}(\mathbf{k})$ is a finite Bloch symbol (Appendix A 2).

For a random lattice at density f_C no translational invariance survives. We treat such a lattice as a mosaic of locally ordered patches, each homogeneous or periodic over a few lattice spacings, so that the local spectral radius Λ_M of a prevalent motif M governs damage growth within its patches. In the thermodynamic limit the onset of global damage spreading is then set by the first statistically prevalent motif to reach marginal stability,

$$s = 1 \iff \Lambda_M = 1 \quad \text{for the first prevalent motif } M, \quad (10)$$

an effective-medium equivalence established in Appendix A 3. It applies in the limit $d_0 \rightarrow 0$ probed by the Derrida slope, the regime in which linear stability analysis is valid [36], and the resulting boundaries agree with the numerical contours of Fig. 3 throughout the parameter range studied.

4. From spectral instability to payoff geometry

In all nonhomogeneous cases we obtain the thresholds not by diagonalizing $\hat{J}(\mathbf{k})$ but directly from payoff balance. At the interface of each characteristic motif there is a pair of competing agents—one cooperator and one defector—whose relative payoffs determine whether the motif grows or shrinks. Marginal stability is reached when the two attain equal payoff, so that neither strategy can invade the other's territory. This *payoff-balance condition* is linear in (u, v) and is the analytical origin of the nearly linear boundaries in Fig. 1. The procedure for each motif is (i) identify the ordered reference pattern it represents, (ii) locate the competing pair at its contested interface, (iii) set their payoffs equal using Eqs. (2) and (3), and (iv) solve for the line in the (u, v) plane. Only two structural properties of the dynamics are used: translational invariance (block-circulant Jacobian) and Boolean, idempotent switching coefficients; the game payoffs enter only through step (iii). The framework therefore applies to any CA sharing these properties.

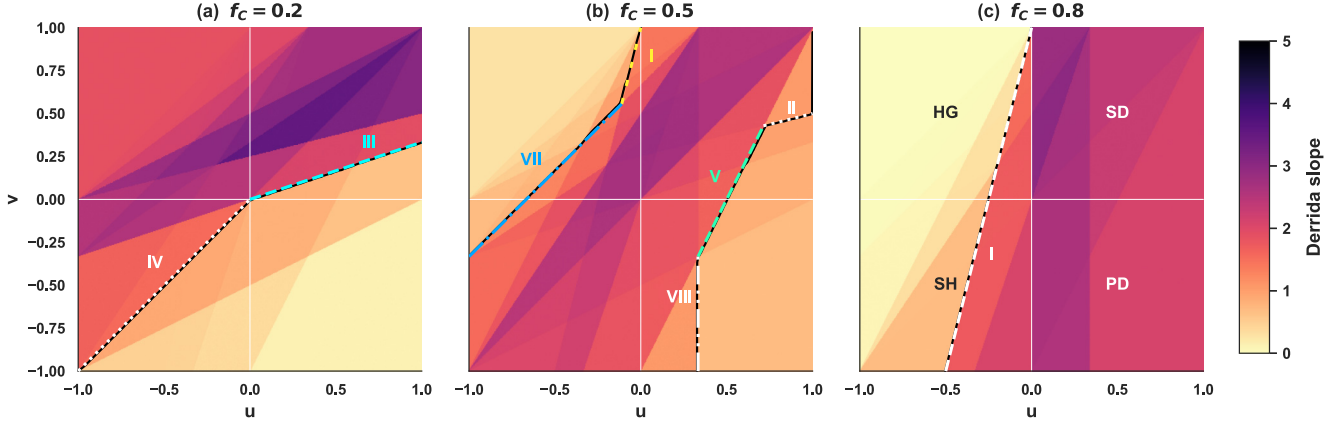


FIG. 3. The analytically derived instability lines (Table I) quantitatively predict the numerical chaos boundaries, with each boundary segment attributable to a single dominant motif. Black contours show the empirical Derrida boundary $s \approx 1$ from Fig. 1; colored dashed lines are the analytical predictions from Table I. Panel (a) ($f_c = 0.2$): the chaotic region is bounded below by the stripe equilibrium line IV ($v = u$) and the nucleation line III ($v = u/3$). Panel (b) ($f_c = 0.5$): the upper boundary of the “chaotic corridor” follows lines VII and I; the lower boundary is a composite of lines V, II, and VIII. Panel (c) ($f_c = 0.8$): dominated by the single-defector invasion line I ($v = 4u + 1$). The shift in dominant line across panels reflects how f_c controls which local geometric motif is statistically prevalent in the reference configuration.

B. Derivation of critical boundaries from motif geometry and marginal stability

We now apply the procedure of Sec. IV A, Step 4, to each characteristic motif: Identify the ordered reference pattern, locate the competing pair at its contested interface, and set their payoffs equal to obtain the marginal-stability line in the (u, v) plane. The microscopic geometries are illustrated in Fig. 2 and all results are collected in Table I.

Several of the payoff-balance conditions derived below have appeared previously as cluster invasion criteria. Hauert [23] showed that a 2×2 cooperator cluster persists when $2 + 2S > T$ and derived analogous inequalities for 1×1 and 3×3 clusters on both von Neumann and Moore neighborhoods; Szabó and Fáth [3] obtained similar thresholds across the literature. These earlier results were obtained by direct score comparison at specific cluster geometries, without connecting those thresholds to spectral stability or chaotic dynamics. The present framework reinterprets each invasion criterion as the payoff-balance condition of a specific motif at marginal stability. To our knowledge, this is the first work to place the known cluster-invasion thresholds within a unified damage-spreading framework and to relate them systematically to the observed chaotic phase boundaries.

The wave vector assigned to each motif is a representative label of the lowest-wave-number periodic background possessing the corresponding symmetry; it organizes the motifs but is not an input to any computation. The thresholds themselves follow from payoff balance at each motif’s contested interface.

1. Long-wavelength sector ($\mathbf{k} \rightarrow (0, 0)$)

The conditions in this sector describe localized nucleation events—a single invader, or the smallest cooperative cluster, embedded in a homogeneous background. For the single-invader cases (Lines I and II) the marginal-stability condition

is read directly from the Boolean derivative: By definition $J_* = 1$ precisely when flipping one site changes the focal update, so the threshold is the locus where J_* switches value—equivalently, where the invader and the adjacent defender attain equal payoff. Line III is the smallest *two-site* cluster (a cooperative pair, each member at $n_C = 1$ through mutual support); being a two-site object it is not a single-site sensitivity but a cluster-nucleation threshold, obtained from the same payoff-balance principle at the pair’s contested interface. None of the three requires a spectral computation.

We retain the label $\mathbf{k} \rightarrow (0, 0)$ only for consistency with the spectral framework: A localized nucleus is a Kronecker delta, and hence broadband in Fourier space, exciting all modes equally. Since $J_0, J_* \geq 0$, the homogeneous dispersion $\lambda(\mathbf{k}) = J_0 + 2J_*[\cos k_x + \cos k_y]$ is maximized at the uniform mode, so the fastest-growing component of that broadband perturbation is the $\mathbf{k} \rightarrow (0, 0)$ sector. The label is therefore an interpretation, not an input to the derivation, and not a claim that the nucleus is a literal $\mathbf{k} = (0, 0)$ eigenmode.

(1) *Defector invading all-C* [Fig. 2(a)]: A single defector surrounded by cooperators has $n_C = 4$, while its nearest cooperator neighbor has $n_C = 3$. From Eqs. (2) and (3), $\Pi_D(4) = 4(1 + u)$ and $\Pi_C(3) = 3 + v$. Setting these equal gives $v = 4u + 1$ (Line I).

(2) *Cooperator invading all-D* [Fig. 2(b)]: A single cooperator surrounded by defectors has $n_C = 0$; its nearest defector neighbor has $n_C = 1$ (one cooperative neighbor from the invading cluster). Then $\Pi_C(0) = 4v$ and $\Pi_D(1) = 1 + u$, giving $v = (u + 1)/4$ (Line II).

(3) *Pair invasion/filament nucleation* [Fig. 2(c)]: A two-site cooperative cluster embedded in a defector background: Each cooperator in the pair has one cooperative neighbor and three defector neighbors ($n_C = 1$), and competes with an adjacent defector that also has $n_C = 1$. Then $\Pi_C(1) = 1 + 3v$ and $\Pi_D(1) = 1 + u$, giving $v = u/3$ (Line III) [37].

2. Stripe modes ($\mathbf{k} = (0, \pi), (\pi, 0)$)

The stripe mode corresponds to alternating columns (or rows) of cooperators and defectors. In this striped background every agent has exactly two cooperative neighbors and two defector neighbors ($n_C = 2$), regardless of their own strategy. This is the *self-dual point* of the dynamics: Cooperators and defectors experience identical neighborhood compositions, so the stripe is neutrally stable—neither strategy has a payoff advantage at the interface. The balance condition is simply $\Pi_C(2) = \Pi_D(2) \Rightarrow \boxed{v = u}$ (Line IV). Since $\Pi_C(2) > \Pi_D(2)$ for $v > u$ and the reverse for $v < u$, the C-D domain wall favors cooperators above this line and defectors below it—domain healing versus roughening in the language of the equivalent ferromagnetic (coordination-game) kinetic-Ising mapping [3].

Transverse displacements of the stripe front create *kinks*—sites where the local neighborhood deviates from the ideal $n_C = 2$ configuration:

(1) *Cooperator kink* [Fig. 2(e)]: a cooperator protruding into defector territory acquires an extra cooperative neighbor ($n_C = 3$) and competes with an adjacent defector still at $n_C = 2$. $\Pi_C(3) = \Pi_D(2) \Rightarrow \boxed{v = 2u - 1}$ (Line V).

(2) *Defector kink* [Fig. 2(f)]: a defector protruding into cooperator territory reduces the cooperative-neighbor count of an adjacent defector to $n_C = 1$, which then competes with a cooperator still at $n_C = 2$. $\Pi_C(2) = \Pi_D(1) \Rightarrow \boxed{v = (u - 1)/2}$ (Line VI).

3. Oblique mode ($\mathbf{k} = (\pi, \pi/2), (\pi/2, \pi)$)

The oblique mode breaks the square-lattice symmetry between horizontal and vertical directions, so its contested interface pairs two agents with markedly different neighborhood compositions. In the configuration of Fig. 2(g), a defector has $n_C = 3$ while the adjacent cooperator has only $n_C = 1$. $\Pi_C(1) = \Pi_D(3) \Rightarrow \boxed{v = u + 2/3}$ (Line VII).

4. Core mode ($\mathbf{k} = (0, \pi/3), (\pi/3, 0)$)

When cooperative clusters are thick enough, interior cooperators acquire four cooperative neighbors ($n_C = 4$) and are entirely shielded from direct interaction with defectors. Instability of the cluster then originates not at the outermost boundary but one layer in, where a “pocket” defector at $n_C = 3$ competes with a fully shielded core cooperator at $n_C = 4$ [Fig. 2(h)]. Since $\Pi_C(4) = 4$ is independent of v , the balance condition involves only u : $\Pi_C(4) = \Pi_D(3) \Rightarrow \boxed{u = 1/3}$ (Line VIII). This is the only vertical boundary line in Table I, reflecting the fact that core erosion is driven purely by the temptation to defect u and is insensitive to the risk of cooperation v .

C. Connecting Derrida slopes with motif prevalence

The analytical lines obtained above and summarized in Table I predict eight candidate boundaries in the (u, v) plane. Which line is observed numerically depends on which motif is most readily excited by the initial configuration—and that depends on the initial cooperator density $f_C(0)$. A random lattice at density $f_C(0)$ predominantly contains the local

motifs that are statistically most likely at that density, and the observed boundary is set by whichever of those motifs first loses marginal stability. We treat three representative densities in turn.

1. Low initial cooperation ($f_C(0) = 0.2$)

When cooperators are rare, the lattice is predominantly defector territory and the statistically prevalent motifs are small cooperative clusters embedded in a defector background. The relevant motifs are therefore those clusters themselves (isolated pairs and single sites), governed by the long-wavelength nucleation lines, together with the C-D domain walls that bound any cooperative cluster, governed by the self-dual stripe condition (Line IV). In the SD region ($u > 0, v > 0$), the dominant boundary is Line III ($v = u/3$, pair nucleation): a two-site cooperative cluster can sustain itself and grow only above this threshold, so the chaotic phase is bounded below by this line. In the SH region ($u < 0, v < 0$) the game is a coordination game with two stable phases—all-C and all-D—and Line IV ($v = u$) is its self-dual coexistence point. Following the kinetic-Ising interpretation of Szabó and Fáth [3], the C-D domain wall carries a surface tension that changes sign across this line: Above it ($v > u$) cooperative domains have positive surface tension and expand, while below it ($v < u$) they contract. At low $f_C(0)$ the lattice is seeded in the all-D basin with sparse cooperative nuclei, so Line IV is the operative boundary because it determines whether those nuclei grow—nucleating the cooperative phase and escaping the all-D basin—or are absorbed. Although labeled the self-dual *stripe* condition in Table I, Line IV operates as the coexistence condition of the bistable Stag-Hunt game at low $f_C(0)$, not as a signature of prevalent stripes.

2. High initial cooperation ($f_C(0) = 0.8$)

When the lattice is nearly all-cooperator, stability is governed by the resilience of the cooperative background against a single invading defector [4]. For sufficiently large u , a lone defector surrounded by cooperators achieves the highest local payoff ($\Pi_D(4) = 4 + 4u$) and is immediately imitated by all four neighbors, triggering a cascade of defector spreading [3,23]. This is precisely the Line I scenario [see Fig. 2(a)]: The Derrida boundary is dominated by $v = 4u + 1$, as confirmed by Fig. 3(c). Because the reference configuration is nearly homogeneous at $f_C(0) = 0.8$, the Boolean Jacobian is close to its all-C form and the exact identity $s = \Lambda$ [Eq. (9)] holds to good accuracy—consistent with the sharp boundary observed numerically. Once Line I is crossed, the Derrida slope exceeds unity and the system may enter the spatiotemporal chaos regime characterized by “kaleidoscope” patterns and traveling waves of defectors [3,23].

3. Balanced initial conditions [$f_C(0) = 0.5$]

At equal initial frequencies, cooperator and defector motifs coexist in comparable proportions, and several motifs approach marginal stability simultaneously, producing the composite “chaotic corridor” visible in Fig. 3(b). The upper boundary of the corridor is defined by whichever of Line I ($v = 4u + 1$, single-defector invasion) and Line VII ($v = u + 2/3$, oblique interface) is first crossed as (u, v) moves

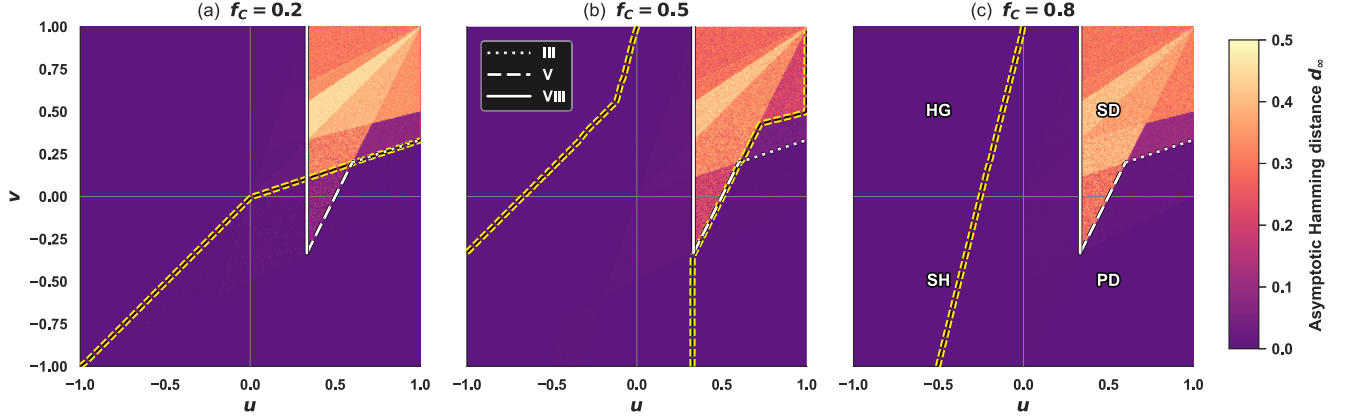


FIG. 4. Two distinct chaos regimes coexist: transient chaos (large) and sustained chaos (smaller interior region), with a subcritical pocket where linear stability and persistent chaos coexist. Each panel shows $d_\infty(u, v)$, the long-run fraction of sites that differ between two initially close replicas, for initial cooperator frequency $f_C(0) = 0.2$ (a), 0.5 (b), and 0.8 (c). Color encodes d_∞ on a perceptually uniform sequential scale: dark purple indicates $d_\infty \approx 0$ (eventual synchronization) and lighter colors indicate $d_\infty > 0$ (sustained chaos). Yellow contours reproduce the empirical Derrida boundary $s \approx 1$ from Fig. 1; white lines bound the region of sustained chaos ($d_\infty > 0$). The gap between the yellow and white lines identifies the *transient chaos* regime: Parameters where $s > 1$ yet $d_\infty \approx 0$, meaning perturbations initially grow but are eventually absorbed as the system organizes into stable geometric domains. The small regions inside the white lines but outside the yellow contour—visible in panels (a) and (b)—are the *subcritical chaos* regime ($s < 1$, $d_\infty > 0$), discussed in Sec. V.

into the chaotic region: These two lines govern the shattering of large uniform strategy blocks into propagating alternating structures. The lower boundary is a composite of Line V ($v = 2u - 1$, cooperator kink), Line II [$v = (u + 1)/4$, cooperator invasion of all- D], and Line VIII ($u = 1/3$, core erosion). Line VIII is the only vertical boundary—a purely u -driven, core-erosion transition: for $u > 1/3$ a pocket defector ($n_C = 3$) outscores even a fully shielded core cooperator ($n_C = 4$), independently of v , so the core site adopts defection at the next update and the shielded interior loses marginal stability. At $f_C(0) = 0.5$ there is sufficient cooperator material for clusters thick enough to develop shielded $n_C = 4$ cores, so this motif is prevalent and its erosion threshold becomes an active boundary—unlike the sparse, thin clusters at low density.

For balanced initial conditions, the reference configuration is maximally disordered, so the exact identity $s = \Lambda$ —which holds only for a homogeneous background—does not apply. Nevertheless, the close agreement between the analytically derived boundaries and the numerical Derrida contours in Fig. 3(b) is consistent with the threshold equivalence [Eq. (10)]: The Derrida slope crosses unity as the most statistically prevalent local motifs—cooperative pairs, stripes, kinks, and cores—successively lose marginal stability, in line with the effective-medium argument of Appendix A 3.

V. ASYMPTOTIC HAMMING DISTANCE AND DYNAMICAL REGIONS

The Derrida slope s is a local quantity: It measures the infinitesimal, one-step growth of damage and therefore characterizes only the *onset* of instability. Whether that instability sustains itself over long times—or is eventually absorbed by the system’s attractor—is measured by the asymptotic

Hamming distance d_∞ [19]. A nonzero d_∞ means that two initially close replicas never synchronize: The system remains in a state of perpetual dynamical activity. By contrast, $d_\infty \approx 0$ means the replicas eventually converge, even if their trajectories diverged transiently. Together, s and d_∞ allow us to distinguish four qualitatively different dynamical regimes, which we now map across the (u, v) plane.

Figure 4 overlays the Derrida boundary ($s \approx 1$, yellow contours) on the d_∞ map, and Fig. 5 shows the resulting four-region cartography. Comparing the two contours reveals a systematic discrepancy: Large regions where $s > 1$ eventually freeze ($d_\infty \approx 0$), identifying a regime of *transient chaos*. In the transient chaos regime, the initial configuration contains locally unstable motifs—so infinitesimal perturbations grow and $s > 1$ —but the game’s global attractor eventually organizes the lattice into stable geometric domains, suppressing all residual damage. A concrete example is the HG region at $f_C(0) = 0.2$: The predominantly defector background generates motifs with $s > 1$, yet cooperation strictly dominates eventually, so the system inevitably converges to the all- C fixed point and freezes.

The same three lines bound the sustained-chaos boundary for all values of $f_C(0)$ which is a nontrivial feature of the cartography. Sustained chaos is an *asymptotic* property ($d_\infty > 0$): Regardless of the initial density, the long-time dynamics generates a common repertoire of local motifs—cooperative pairs, stripe kinks, and cluster cores—through the continual formation and break-up of clusters, so the boundary is controlled by the same motif-instability lines (III, V, VIII) at every density. Notably, cooperative cores are essentially absent in the initial low- $f_C(0)$ lattice yet form dynamically as clusters nucleate and grow, which is why Line VIII (core erosion) bounds sustained chaos even at $f_C(0) = 0.2$. The boundaries of *sustained* chaos ($d_\infty > 0$) are consistent across all three initial conditions and coincide with specific lines from Table I:

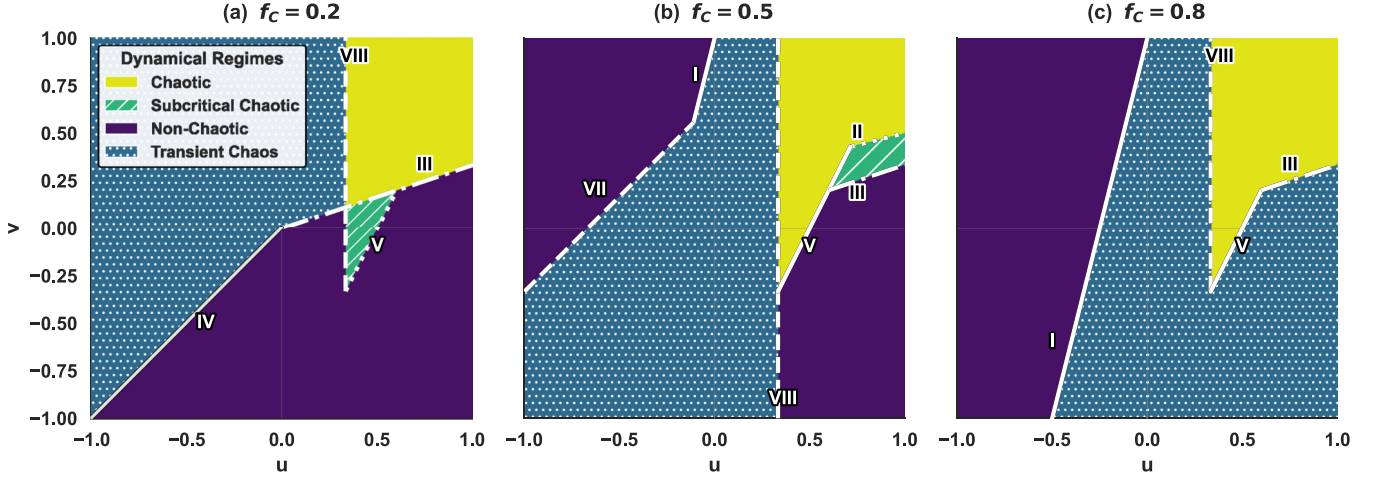


FIG. 5. Three motif instability lines (III, V, VIII) universally bound the sustained-chaos region across all initial cooperator densities, indicating the spectral origin of the phase boundaries. Each point in the (u, v) plane is classified by the joint values of s and d_∞ into one of four dynamical regimes. Color key: Dark blue (ordered, $s < 1$, $d_\infty \approx 0$); dotted light blue (transient chaos, $s > 1$, $d_\infty \approx 0$); yellow (sustained chaos, $s > 1$, $d_\infty > 0$); dashed-green (subcritical chaos, $s < 1$, $d_\infty > 0$). Dashed lines are the analytical instability lines from Table I; the boundaries between colored regions coincide with Lines III, V, and VIII across all three panels. Panels show $f_C(0) = 0.2$ (a), 0.5 (b), and 0.8 (c).

(1) Nucleation limit (Line III, $v = u/3$): For $u > 0.6$, sustained activity requires the pair-nucleation threshold to be exceeded; below this line cooperative clusters cannot self-sustain and the dynamics freezes.

(2) Stripe unzipping limit (Line V, $v = 2u - 1$): In the SD and PD regions, sustained chaos is confined above $v = 2u - 1$ for $u \in (1/3, 3/5)$; below this line, domain walls lock into stable stripe patterns and damage cannot propagate indefinitely.

(3) Core erosion limit (Line VIII, $u = 1/3$): The vertical line $u = 1/3$ marks a universal transition between transient and sustained chaos across all three initial cooperator frequencies, reflecting the purely u -driven nature of cluster-core instability.

Figures 4(a) and 4(b) further reveal a *subcritical chaotic phase*: regions where $s < 1$ yet $d_\infty > 0$. Specifically, the triangular region bounded by Lines III, V, and VIII in Fig. 4(a), and the region bounded by Lines II, III, and V in Fig. 4(b), exhibit sustained chaos that the Derrida slope alone cannot predict. The mechanism is nonlinear: infinitesimal perturbations decay ($s < 1$), so the linearized framework predicts stability, but the basin of attraction of the ordered state is bounded. A finite-amplitude perturbation—one large enough to push the system across the basin boundary—can reach a self-sustaining chaotic attractor that the linear analysis cannot see. Simulations on lattices up to $L = 200$ and runs extended to $t = 10^5$ time steps indicate that $d_\infty > 0$ in these regions is not a finite-size or finite-time artifact. Figure 5 shows the complete cartography.

This subcritical regime parallels the subcritical transition to turbulence in shear flows, where the laminar base state is linearly stable yet finite-amplitude perturbations can sustain turbulence [38]. In both settings the ordered (laminar) state is linearly stable while finite-amplitude perturbations sustain chaotic (turbulent) activity. We draw the analogy at the level of this shared phenomenology; a quantitative characterization of the basin boundary and any critical perturbation amplitude is left to future work.

VI. CONCLUSION

This paper asked a simple question: *Where do the chaotic phase boundaries in spatial evolutionary games lie and why?* The answer is that each boundary marks the loss of marginal stability of a characteristic spatial motif. Each threshold is located by a payoff-balance condition at the motif's contested interface and interpreted spectrally as the loss of stability of the corresponding ordered background. The framework derives each of these boundaries from a single microscopic instability mechanism. The boundaries are sharp and approximately linear, and their positions shift systematically with the initial cooperator density (Sec. III)—features that were previously only observed numerically. This supplies the analytical account of the spatial chaos first demonstrated by Nowak and May [1].

Our central result is that motif marginal stability sets the chaotic phase boundaries. Applying an effective-medium argument—the random lattice treated as a mosaic of locally ordered patches—the onset of damage spreading is controlled by the first statistically prevalent local motif to reach its threshold ($\Lambda_M = 1$). This is what places the analytical lines on top of the numerical chaos boundaries. The homogeneous identity $s = \Lambda$ anchors the spectral interpretation: In this exactly solvable case, the Derrida slope and the spectral radius of the Boolean Jacobian coincide. This correspondence addresses the explanatory gap identified in the introduction. It accounts not only for *that* chaos occurs but also for the *family of motifs* that can trigger it, *where* in parameter space the transition happens, and *why* the dominant motif shifts with the initial cooperator density. It also supplies the mechanism behind a previously unmade connection: The cluster-invasion thresholds of Hauert [23] and Szabó and Fáth [3] coincide with the payoff-balance conditions of specific motifs at marginal stability. The central outcome is not the recovery of any individual boundary. It is the organization of the entire dynamical phase diagram by the motif-instability lines of Table I. The linear-analysis lines set the onset of chaos

($s \approx 1$). A small, fixed subset (notably III, V, and VIII) then bounds the *sustained*-chaos region across all densities studied. The same few lines govern sustained chaos independently of cooperator density.

We further identify a subcritical chaotic regime ($s < 1$, $d_\infty > 0$), in which finite-amplitude perturbations sustain chaos despite linear stability. This is the spatial evolutionary game analog of the subcritical transition to turbulence in shear flows [38]. Within this regime the ordered state is *metastable*: Robust to infinitesimal perturbations but vulnerable to sufficiently large structural fluctuations.

Several natural extensions remain open. First, asynchronous updating and stochastic imitation rules break the synchronous, deterministic one-step structure on which the block-circulant form of the Boolean Jacobian relies. Stochastic noise is then expected to smear the sharp spectral boundaries into crossover regions [39]. Second, extending the framework to heterogeneous topologies requires replacing the DFT with the graph Fourier transform, whose eigenbasis is set by the graph Laplacian rather than by translational symmetry. Third, conformity dynamics create effective surface tension and smooth cooperative boundaries [40]. These may interact nontrivially with the motif instabilities identified here, potentially shifting or splitting the critical lines. Each extension preserves the core logic of the framework—linearize, decompose, find the marginal-stability condition—while replacing the specific technical ingredients. The present results therefore serve as the analytically tractable baseline to which these additional effects can be added systematically.

Beyond evolutionary games, the ingredients of this analysis are generic. The exact identity $s = \Lambda$ for a homogeneous background, and the motif-threshold equivalence for disordered ones, require only a deterministic, translationally invariant update rule with Boolean (idempotent) switching coefficients. The linearized operator is then block-circulant on each ordered patch [42], and the scalar idempotency argument applies locally (Appendix A 1); the damage-spreading spectral equivalence is generic for this class of CA [31,41]. None of this depends on the payoff interpretation: What matters is the local block-circulant structure of the linearized operator and the idempotency of the Boolean coefficients. The link between damage spreading and spectral instability is therefore a structural feature of this class of cellular automata, not specific to evolutionary games.

ACKNOWLEDGMENTS

The author thanks the two anonymous referees for their careful reading and constructive comments, which substantially improved the manuscript. An AI language model was used for limited language editing of an earlier draft; the present version was revised by the author, who takes full responsibility for the scientific content.

DATA AVAILABILITY

The data, code, and materials supporting the findings of this article are available at Ref. [43].

APPENDIX: THE DERRIDA–SPECTRAL CORRESPONDENCE

This appendix collects the technical material supporting Sec. IV A: the derivation of the exact homogeneous identity $s = \Lambda$ (Appendix A 1), the Bloch structure of the Boolean Jacobian for periodic backgrounds (Appendix A 2), and the effective-medium threshold argument for random configurations (Appendix A 3).

Throughout we use one structural fact. For a homogeneous reference background, the Boolean Jacobian is translationally invariant and therefore block circulant; it is diagonalized exactly by the discrete Fourier transform [42]. For a periodic background, the Jacobian inherits the periodicity of the unit cell and admits a Bloch-Fourier decomposition [35], leading to a finite Bloch symbol $\hat{J}(\mathbf{k})$. For a random configuration this structure holds only locally, motivating the effective-medium argument of Appendix A 3.

1. The exact identity $s = \Lambda$ for homogeneous backgrounds

A spatially random perturbation with flat power spectrum excites all Fourier modes equally. The one-step growth of the total damage—measured by the squared norm $\|\delta\|^2$, which equals the number of damaged sites for Boolean δ —is therefore an average over all modes, weighted by their squared amplification factors $|\lambda(\mathbf{k})|^2$. For a general linear operator this mode average and the spectral radius $\Lambda = \max_{\mathbf{k}} |\lambda(\mathbf{k})|$ are unrelated quantities. For the homogeneous Boolean Jacobian they coincide, forced together by the idempotency of the coefficients, $J_0^2 = J_0$ and $J_*^2 = J_*$.

Expand $|\lambda(\mathbf{k})|^2 = (J_0 + 2J_*[\cos k_x + \cos k_y])^2$ and average over the Brillouin zone. The cross terms between J_0 and the cosines vanish by symmetry, as does the mixed term $\langle \cos k_x \cos k_y \rangle = 0$; each cosine-squared term averages to $\langle \cos^2 k_x \rangle = \frac{1}{2}$. The zone average is therefore

$$\langle |\lambda(\mathbf{k})|^2 \rangle = J_0^2 + 4J_*^2 = J_0 + 4J_*, \quad (\text{A1})$$

where the second equality applies idempotency. Since $J_0, J_* \geq 0$, the maximum of $|\lambda(\mathbf{k})|$ is attained at the uniform mode $\mathbf{k} = (0, 0)$, where $\cos k_x + \cos k_y$ peaks, and equals $J_0 + 4J_*$ as well. Because Boolean damage satisfies $\|\delta\|^2 = L^2 d$, the zone-averaged mean-square amplification [Eq. (A1)] is precisely the Derrida slope, and

$$s = \Lambda = J_0 + 4J_*. \quad (\text{A2})$$

This establishes Eq. (9) of the main text. The argument uses only translational invariance (so that plane waves diagonalize the Jacobian) and idempotency of the Boolean coefficients; it is therefore a structural property of this class of cellular automata rather than a feature of the game-theoretic payoffs.

2. Bloch structure for periodic backgrounds

For a background with a nontrivial unit cell—an alternating stripe or a cooperative core—the payoff ordering varies between site classes within the cell, so the Boolean derivative is no longer the same at every site. A cooperator in the interior of a cluster ($n_c = 4$) has a different switching sensitivity than one at the cluster boundary ($n_c = 2$ or 3), and the coefficient for a north neighbor may differ from that for an east neighbor

when the background breaks the four-fold rotational symmetry. The linearized rule therefore carries one coefficient set per site class,

$$\delta_{n,m}(t+1) = \sum_{(r,s) \in \mathcal{N}_0} J_{r,s}^{(\alpha(n,m))} \delta_{n+r,m+s}(t), \quad (\text{A3})$$

where $\alpha(n, m)$ identifies the site class at (n, m) . The coefficients are not invariant under single-site translations, but they are invariant under translations by a full unit cell (e.g., by two sites for an alternating stripe). Grouping sites into unit cells, the Jacobian is circulant at the level of cells, with each entry a small matrix acting on the site-class indices—that is, block-circulant. It is diagonalized by a Bloch-Fourier transform, with eigenvalues given by those of a finite Bloch symbol $\hat{J}(\mathbf{k})$ whose dimension equals the number of site classes, and the marginal-stability criterion is $\rho(\hat{J}(\mathbf{k})) = 1$.

Because the Boolean coefficients are piecewise constant in (u, v) and change only when a local payoff ordering flips, the Bloch symbol is likewise piecewise constant within each payoff-ordering region. The loci at which its stability properties can change are therefore exactly the payoff-balance lines of the corresponding motifs. This is why, in the main text, the instability thresholds are obtained directly from payoff balance at each motif's contested interface (Sec. IV B) rather than by explicit diagonalization of $\hat{J}(\mathbf{k})$; the resulting boundaries are verified against the numerical Derrida contours in Fig. 3.

We note that for these multiclass symbols the scalar idempotency argument of Appendix A 1 does not apply, and the identity $s = \Lambda$ is not expected to hold exactly; the periodic motifs are treated at the threshold level only.

3. Threshold equivalence for random configurations

When the reference configuration is drawn randomly at density f_C , the Boolean Jacobian $\mathbf{J}(S^0)$ depends on the realization S^0 and is no longer globally block circulant, so the equality $s = \Lambda$ cannot be expected to hold pointwise. We

establish the threshold equivalence (10) through an effective-medium argument, valid in the thermodynamic limit, where $\Lambda_M = \max_{\mathbf{k}} \rho(\hat{J}_M(\mathbf{k}))$ is the spectral radius of the (possibly multiband) Bloch symbol of motif M (Appendix A 2).

i. Heuristic physical motivation. The equivalence (10) is not a rigorous mathematical proof but an effective-medium estimate. A random configuration can be viewed as a spatial superposition of local geometries (single invaders, pairs, stripes, kinks, cores) whose relative frequencies are determined by $f_C(0)$. Each geometry possesses an associated local instability threshold Λ_M . When all statistically prevalent motifs are stable ($\Lambda_M < 1$), perturbations fail to grow in the environments that dominate the lattice, and the ensemble-averaged damage contracts ($s < 1$). Conversely, once a statistically prevalent motif becomes unstable ($\Lambda_M > 1$), perturbations are linearly amplified in a finite fraction of local environments across the lattice, leading to macroscopic growth of the ensemble-averaged damage ($s > 1$).

ii. Consequence. Within this picture, the transition $s = 1$ occurs when the first statistically prevalent local geometry loses marginal stability [41], which explains the close agreement between the analytically derived boundaries (Table I) and the numerically measured Derrida contours (Fig. 3), even for maximally disordered initial conditions ($f_C(0) = 0.5$).

iii. Remark on the magnitude of s in the chaotic interior. Deep inside the chaotic phase, the numerical Derrida slope and the spectral radius of any single ordered motif may differ quantitatively, because the random Jacobian mixes contributions from multiple local geometries with distinct instability spectra. The threshold correspondence (10) depends only on the *sign* of the linear growth rate and is insensitive to such interior deviations.

iv. Scope. The arguments above address only the linear regime ($d_0 \rightarrow 0$). Chaotic dynamics may persist even where $s < 1$, driven by finite-amplitude perturbations rather than linear instability (Sec. V); this subcritical regime lies outside the scope of the spectral ansatz and is characterized numerically through d_∞ .

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